Milling Machine

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Milling machine

What it is used for

A milling machine removes material from a workpiece using a rotating cutting tool. It produces slots, flat surfaces, grooves and other shapes in metal or suitable materials.

Main parts

- Identify the machine base, column and worktable supporting the workpiece.
- Locate the spindle that holds and rotates the milling cutter.
- Identify the milling cutter, tool holder and spindle locking arrangement.
- Locate the table movement controls for longitudinal, cross and vertical travel.
- Identify the workholding vice, clamps and T-slots on the table.
- Locate the speed selector, feed controls and emergency stop.
- Identify protective guards, coolant equipment and lubrication points, where fitted.

Before you start

1. Obtain teacher supervision and review the machine operating instructions.
2. Wear safety goggles and suitable workshop clothing; secure long hair.
3. Remove jewellery and avoid loose clothing near rotating components.
4. Inspect the cutter, tool holder, guards and electrical connections for damage.
5. Secure the workpiece firmly in a vice or with suitable clamps.
6. Check cutter clearance, spindle speed, feed setting and workpiece alignment.
7. Remove spanners and loose objects from the table before starting.

How to use

1. Select a cutter suitable for the material and required operation.
2. Switch off the machine before installing or changing the cutter.
3. Secure the cutter and workpiece using the correct holding arrangements.
4. Set the appropriate spindle speed, feed rate and cutting depth.
5. Position the cutter and workpiece with the machine stopped where required.
6. Close the guards and ensure tools are clear of moving parts.
7. Start the spindle and allow it to reach the selected speed.
8. Engage the feed gradually while monitoring cutting conditions.
9. Use approved coolant where required and keep hands away from the cutter.
10. Stop the machine before measuring, adjusting or removing the workpiece.

⚠ Safety

- Never touch the rotating cutter or remove chips by hand.
- Never wear gloves near rotating cutters or exposed spindle components.
- Keep guards in place and never bypass safety interlocks.
- Use a brush or chip hook only after the machine stops.
- Do not measure or adjust the workpiece while the cutter rotates.
- Switch off and isolate power before changing cutters or clearing jams.

Care & storage

- Remove chips with a brush after switching off and isolating power.
- Clean the table, vice and exposed machine surfaces after use.
- Lubricate designated points and maintain coolant according to instructions.
- Inspect cutters, belts and guards regularly and report any damage.

Common mistakes to avoid

- Clamping the workpiece loosely, allowing it to shift during cutting.
- Selecting an unsuitable speed or feed for the cutter and material.
- Measuring the workpiece while the spindle or cutter is moving.
- Removing metal chips with bare hands or while the machine runs.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar